Smaller Is Not Always Faster: The Memory–Performance Trade-off of LLM Quantization on an RTX 5090

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Abstract

Quantization stores the weights of a large language model (LLM) in 8 or 4 bits instead of 16, and formats are usually compared by this bit-width. We show that the bit-width alone predicts neither the memory a format saves, nor its speed, nor its cost in quality. On one NVIDIA RTX 5090 we measure eight configurations on two inference stacks (Hugging Face Transformers with bitsandbytes, and vLLM) for five models from 3B to 32B parameters, with 1,890 timed runs, real chat traffic and three quality benchmarks. The results form a chain. First, 4-bit weights shrink a 7B model only 2.60×–2.70×, not 4×, because the embedding, the KV cache and the activations stay in 16 bits. Second, the saved bytes become speed only when decoding is memory-bound and a fused kernel hides the dequantization: at batch size 1, 4-bit weights decode 2.44× faster on vLLM but only 1.13× faster with bitsandbytes. Third, batching makes decoding compute-bound, so these weight-only gains vanish, as a roofline model predicts; only FP8, whose fused kernel also does the multiply in 8 bits, stays faster at every batch size. Fourth, the gains cost little quality at 8 bits and 0.6–2.8 MMLU points at 4 bits, more on smaller models. The best use of the saved memory is therefore a larger model: Qwen3-32B in 4 bits scores 78.1% on MMLU against 69.6% for Qwen3-8B in BF16, with only 1.20× the weight memory.

1 Introduction

Large language models (LLMs) need a lot of GPU memory. The 16-bit weights of a 7B-parameter model take 15 GB, and the key–value (KV) cache of every running request comes on top. To generate each token, the GPU reads all the weights from memory, so generation is often limited by memory bandwidth rather than by arithmetic [9, 22]. Quantization stores the weights, and sometimes the activations, in 8 or 4 bits. Formats are named by what they quantize and to how many bits (W4A16 means 4-bit weights and 16-bit activations), which suggests a simple rule: fewer bits mean less memory, faster generation and a small loss in quality.

This rule can fail at each of its three steps. *Memory:* the embedding, the output layer, the KV cache and the activations stay in 16 bits, so the saving is smaller than the bit-width suggests. *Speed:* most formats must convert the weights back to 16 bits (dequantize them) before each matrix multiply. This is free only if a fast kernel hides it while the GPU waits for memory, and large batches make the GPU wait for arithmetic instead. *Quality:* the rounding error depends on the method and on the model.

Method papers report accuracy and kernel speed separately, usually on data-center GPUs [7, 8, 17, 18]; comparison studies focus on accuracy [13, 16] or on one serving stack [14]. But a user with one

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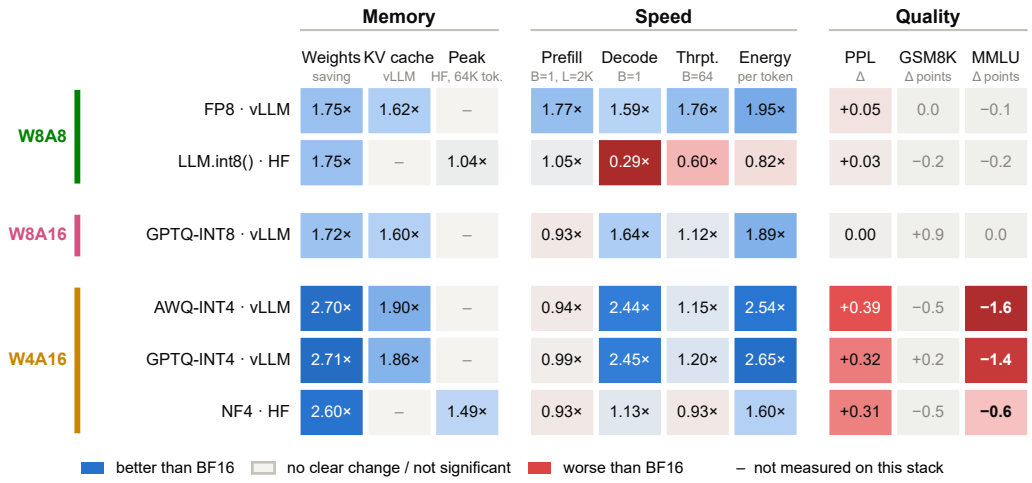


Figure 1: **What each quantization configuration does to each metric** (Qwen2.5-7B). Rows group the configurations by what they quantize; each cell compares one with BF16 on the same stack. Memory and speed are ratios (> 1: better); quality cells are coloured only if significant (Section 3). The two W8A8 rows differ because of their kernels (Section 4.4).

consumer GPU also picks a library, and the same format runs on very different kernels in Hugging Face Transformers (bitsandbytes) and in vLLM (Marlin, CUTLASS).

We measure the whole trade-off on an NVIDIA RTX 5090, a 32 GB consumer GPU, and answer the four questions of our proposal. **Q1:** How much memory does quantization really save? **Q2:** How do the time to first token (TTFT), the time per output token (TPOT) and the throughput change? **Q3:** At which batch sizes and prompt lengths does a low-bit format become faster, or slower, than BF16? **Q4:** How much quality does it cost? We test six quantized configurations and BF16 on two stacks and five models (Qwen2.5 at 3B, 7B and 14B; Qwen3-8B and Qwen3-32B to check generality). Figure 1 shows the effect of every configuration on every metric. Its columns give our main findings, and each finding limits the next:

1. **Memory (Q1, Section 4.1).** 4-bit weights shrink Qwen2.5-7B by only 2.60×–2.70×, not 4×, and less at run time; the saving pays off as capacity.
2. **Speed (Q2, Section 4.2).** The saved bytes become speed only with a fused kernel: vLLM turns 90–95% of them into speed at batch size 1, bitsandbytes 44% or less.
3. **Crossover (Q3, Sections 4.3 and 4.4).** The gain lasts only while decoding is memory-bound, so batching ends it ($B \approx 175$); only FP8, whose fused kernel also multiplies in 8 bits, stays.
4. **Quality (Q4, Sections 4.6 and 4.7).** The cost is small at 8 bits and 0.6–2.8 MMLU points at 4 bits, less than the gain from a larger model.

We release the harness, the raw data and the scripts that produce every number, table and figure in this report.[1]

2 Background and related work

Prefill and decode. An LLM serves a request in two phases. *Prefill* processes all L prompt tokens at once, so for a batch of B requests each linear layer multiplies its weights by $M = B \cdot L$ rows; prefill sets the TTFT. *Decode* then generates one token per request per step, so each layer sees only $M = B$ rows; one step takes one TPOT. The keys and values of earlier tokens are kept in the KV cache.

[1] https://github.com/zz9tf/6359-SYS-ARCHITECT-course-work

A roofline view. The roofline model [25] tells which resource limits a kernel. For a linear layer $y = xW^\top$ with $x \in \mathbb{R}^{M\times K}$, $W \in \mathbb{R}^{N\times K}$ and b bytes per weight, the arithmetic intensity (operations per byte moved) is

$$I(M) \;=\; \frac{2MNK}{b\,NK + 2MK + 2MN} \;\approx\; \frac{2M}{b} \qquad (M \ll N, K). \tag{1}$$

The kernel is memory-bound when $I(M)$ is below the ridge point R (peak FLOP/s divided by memory bandwidth) and compute-bound above it. On our GPU, $R \approx 135$ FLOP/byte for BF16. Equation (1) gives three predictions, which we test in Section 4:

- **P1** Small-batch decode is memory-bound, so fewer weight bytes should give a proportional speedup, if the kernel hides the dequantization.
- **P2** Weight-only formats (W8A16, W4A16) still compute in BF16: their speedup shrinks once they reach the compute roof ($M \approx bR/2$) and is gone once BF16 is compute-bound too ($M \gtrsim R$). W8A8 formats use 8-bit tensor cores, at least twice as fast, so they stay faster.
- **P3** Prefill is compute-bound, so only W8A8 formats can shorten the TTFT.

The roofline leaves out the CPU time needed to launch each kernel. If this time exceeds the GPU time, the step is *host-bound*, and no saving of bytes helps.

Formats. LLM.int8() [5] (INT8 weights and activations, with large activation columns kept in FP16) and NF4 [6] (4-bit NormalFloat weights) come from bitsandbytes and run in Hugging Face Transformers [12]. GPTQ [7] and AWQ [17] choose the rounding with second-order or activation information and store INT4 or INT8 weights; vLLM [15] runs them with Marlin kernels [8]. FP8 [21] stores weights and activations in an 8-bit float that recent GPUs support in hardware: the floating-point form of W8A8, which SmoothQuant [26] enables for INT8.

Related work. QServe [18] notes that W4A16 kernels lose their advantage in large-batch serving; other work quantizes activations to 4 bits [2] or compresses the KV cache [11, 19]. Dettmers and Zettlemoyer [4] show that 4-bit models give the best accuracy for a fixed number of model bits. Jin et al. [13] and Li et al. [16] compare methods on accuracy, and Jin et al. note that quantization can slow inference down. Closest to us, Kurtic et al. [14] find on data-center GPUs that W4A16 suits latency-bound use and W8A8 suits high-throughput serving. We confirm this on a consumer GPU and add the Hugging Face path, memory at run time, and a kernel-level explanation of every crossover.

3 Experimental setup

Platform. We use one NVIDIA GeForce RTX 5090 (Blackwell, 170 SMs, 32 GB GDDR7, 575 W) in a workstation with an Intel Core Ultra 9 285K and 64 GB of RAM, running Ubuntu 24.04, PyTorch 2.13 (CUDA 13.0), Transformers 5.19, bitsandbytes 0.50.2 and vLLM 0.32 (CUDA graphs on). Our microbenchmarks measure 1.61 TB/s of memory bandwidth and 218 (BF16) and 645 (FP8) TFLOP/s for large matrix multiplies, which puts the BF16 ridge point at about 135 FLOP/byte.

Models and configurations. The main study uses instruction-tuned Qwen2.5 [23] at 3B, 7B and 14B parameters; Qwen3-8B and Qwen3-32B [24] test generality. Table 1 arranges the configurations by two choices: what is quantized, and which stack runs it. The Hugging Face (HF) stack is what a user gets with one extra argument to `from_pretrained`: bitsandbytes quantizes the weights while loading, and PyTorch runs the model step by step. The vLLM stack is a serving engine with fused kernels and CUDA graphs. In every configuration the embedding, the output layer (LM head) and the norms stay in 16 bits.

Workloads. *Controlled sweep:* each configuration generates exactly 128 tokens for $B \in \{1, 2, 4, \ldots, 64\}$ prompts (up to 256 on vLLM) of $L \in \{128, 512, 1024, 2048\}$ random tokens, so every configuration does the same work; a smaller sweep varies the output length. *Real traffic:* 300 ShareGPT [1] conversations (median prompt 31 tokens, median answer 284 tokens), served by vLLM to 1–64 concurrent clients and by HF in static batches of 16. *Quality:* WikiText-2 [20] perplexity on 2048-token windows; GSM8K [3] on all 1,319 test questions with zero-shot chain of thought and greedy decoding; MMLU [10] on all 14,042 test questions, zero-shot. Prompts use each model's chat template, and Qwen3 runs in non-thinking mode.

Table 1: The six quantized configurations by what they quantize (columns) and the stack that runs them (rows); both stacks also run BF16. Figures use these colours; vLLM is solid/filled, HF dashed/hollow.

Stack	**W8A8** 8-bit weights + activations	**W8A16** 8-bit weights	**W4A16** 4-bit weights
HF + bitsandbytes	LLM.int8()†	—	NF4
vLLM + fused kernels	FP8 (CUTLASS)	GPTQ-INT8 (Marlin)	AWQ-, GPTQ-INT4 (Marlin)

†INT8, with activation columns above 6.0 kept in FP16. bitsandbytes (NF4: block 64) and FP8 (E4M3) quantize while loading; GPTQ and AWQ use the official Qwen checkpoints (group size 128).

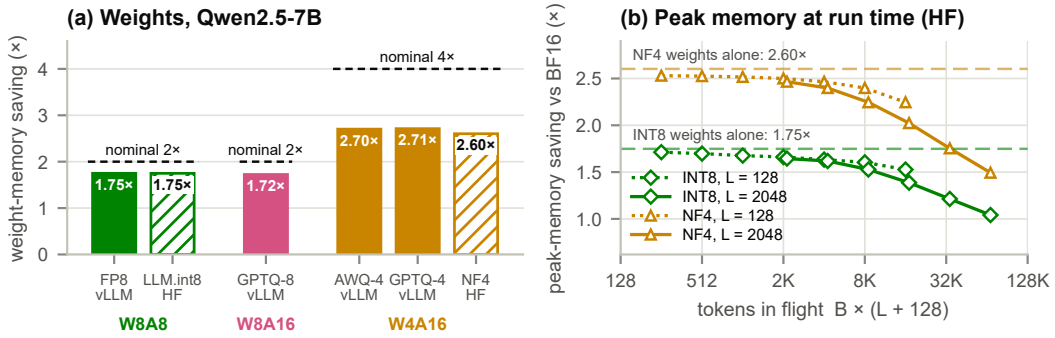


Figure 2: **Q1: the saving is smaller than the bit-width suggests.** (a) Weight-memory saving of each configuration on Qwen2.5-7B against the nominal ratio of its bit-width (dashed). (b) Saving of the peak memory on HF as more tokens are in flight; dashed lines give the saving of the weights alone.

Metrics and statistics. TTFT is the time to the first output token, TPOT the mean time of each later token, and throughput the number of output tokens per second. Each cell has one warm-up and three measured runs with fresh prompts; the run-to-run variation of the end-to-end time is below 0.7% in 95% of the cells and 2.3% at most. Memory and energy come from PyTorch and NVML. For quality, all configurations answer the same questions, so we compare each with HF BF16 by an exact McNemar test, with a Bonferroni correction over the 18 comparisons of each benchmark ($\alpha = 0.0028$).

4 Results

We explain Figure 1 in the order its effects build on each other: saved bytes (Section 4.1), speed at small batches (Section 4.2), how long it lasts (Sections 4.3 and 4.4), real traffic (Section 4.5), quality (Section 4.6) and generality (Section 4.7).

4.1 Q1: How much memory does quantization actually save?

Answer: *Less than the bit-width suggests, and less again at run time. The saved memory pays off as capacity.*

Weights. Quantization shrinks only the linear layers inside the decoder. The embedding and the LM head, 14% of Qwen2.5-7B's parameters, stay in 16 bits, and each format adds its scales. So 4-bit weights save 2.60×–2.70× instead of 4×, and 8-bit weights 1.75× instead of 2× (Figure 2a). A model that counts these parts predicts every measurement within 0.6% (Appendix A).

Run time. At run time the saving shrinks further, because the KV cache and the activations are not quantized at all. Their share grows with the number of tokens in flight. NF4's saving therefore falls from 2.53× for one short request to 1.49× for 32 requests of 2048 tokens, and LLM.int8(), which keeps extra copies of the activations, falls to 1.04× (Figure 2b).

Capacity. The saved memory is still valuable, but as capacity rather than as a smaller footprint. vLLM turns it into KV cache, so AWQ-INT4 lets Qwen2.5-7B hold 1.90× as many tokens as BF16

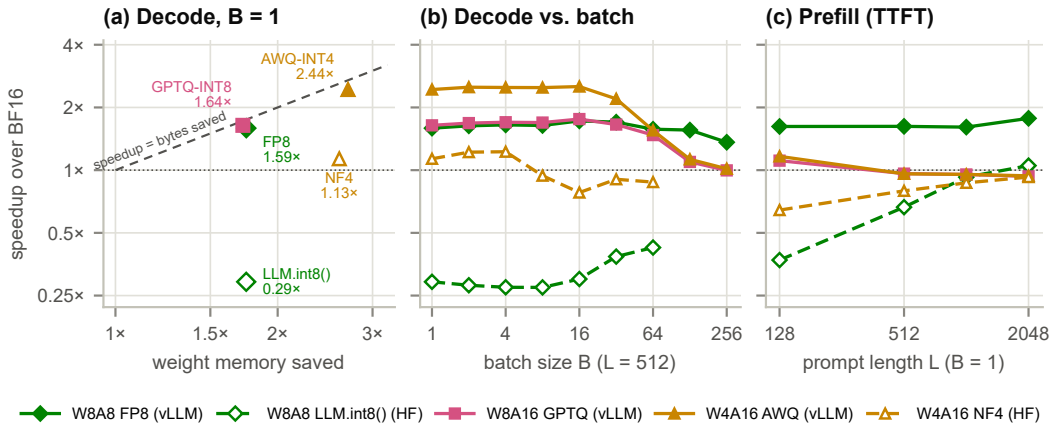


Figure 3: **Q2 and Q3: when the saved bytes become speed** (Qwen2.5-7B, speedup over BF16 on the same stack). (a) Decode speedup at batch size 1 against the weight memory saved: if decode time were set only by the bytes read, every point would lie on the dashed line. (b) Decode speedup as the batch grows (512-token prompts). (c) Prefill (TTFT) speedup for one prompt of growing length.

(355K vs. 187K). The 14B model is the extreme case. In BF16 its 29.5 GB of weights leave no room for a KV cache, even when vLLM may use 95% of the GPU, so vLLM cannot serve it at all. In AWQ-INT4 it runs with 82K cached tokens.

4.2 Q2: How do latency and throughput change?

Answer: *The saved bytes speed up decoding only when a fused kernel hides the dequantization. Prefill gets faster only when the activations are quantized too.*

Decode. At batch size 1, decoding is memory-bound: vLLM's BF16 step takes 9.52 ms, within 99% of the time needed to read the weights once (prediction P1). Fewer weight bytes should therefore shorten the step in proportion. On vLLM they do: AWQ-INT4 decodes 2.44× faster, GPTQ-INT8 1.64× and FP8 1.59×, which is 90–95% of their byte savings (Figure 3a). On HF they do not: NF4 saves almost as many bytes as AWQ-INT4 but is only 1.13× faster, and LLM.int8() is 3.4× slower. The same bit-width can thus speed decoding up or slow it down. The difference lies in the whole stack (method, activation path and engine), which Section 4.4 takes apart.

Prefill. Prefill behaves differently, because it is compute-bound from the start (P3). Fewer weight bytes therefore barely help: AWQ and GPTQ change the TTFT by at most 7%, and NF4 even slows it by 1.26×, since bitsandbytes dequantizes each matrix before using it (Figure 3c). Only the W8A8 formats, which also quantize the activations, shorten prefill: FP8 by 1.77× for a 2048-token prompt, and LLM.int8() by only 1.05×, as its overheads eat most of the gain.

4.3 Q3: When does lower precision pay off?

Answer: *Weight-only gains last only while decoding is memory-bound. FP8 pays off at every batch size.*

The batch-1 gains rest on decoding being memory-bound, and batching removes that condition. Every extra request reuses the same weights, so the work per byte grows with the batch (Equation (1)). The speedup of weight-only formats therefore shrinks as the batch grows (Figure 3b) and is gone near the BF16 ridge point (P2): AWQ-INT4 stops winning at $B \approx 175$ for 512-token prompts (Figure 4). Long prompts add compute-bound prefill to each request, so the gain fades sooner. Long answers add memory-bound decode and raise it (at $B = 16$, 1.26× for 32 output tokens, 2.16× for 512; Appendix B). FP8 does not fade, because its kernel also does the multiply in 8 bits, on tensor cores three times faster than BF16: it stays 1.59×–1.80× faster in every cell. LLM.int8() is W8A8 too, yet it is slower in every cell (0.27×–0.71×), and NF4 wins only below $B \approx 7$. The bitsandbytes formats never reach the memory-bound gain in the first place; Section 4.4 shows why.

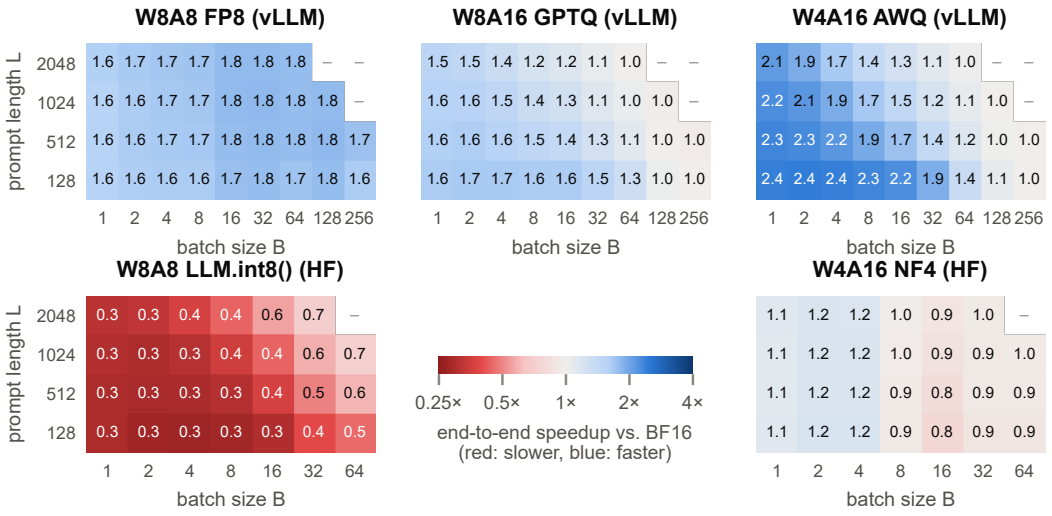


Figure 4: **Q3: where each configuration pays off.** End-to-end speedup of B requests (L-token prompts, 128 generated tokens) over BF16 on the same stack, Qwen2.5-7B. Columns follow the quantization type, rows the stack (no W8A16 runs on HF). Dashes: out of memory or not run. GPTQ-INT4 behaves like AWQ-INT4 (Table 6).

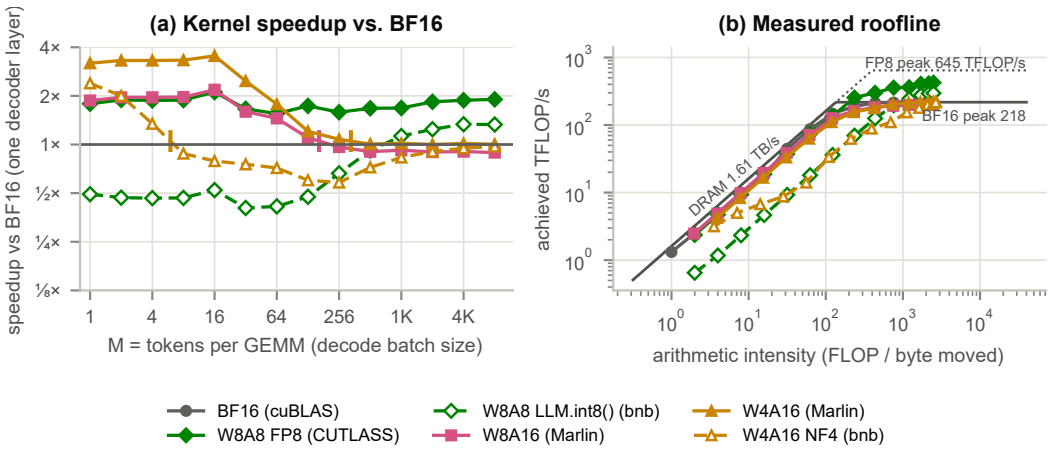


Figure 5: **Why the crossover happens.** (a) Speedup of each low-bit kernel over BF16 cuBLAS on the matrix shapes of one Qwen2.5-7B decoder layer, against the tokens M per multiply (the decode batch size); ticks mark where it falls below 1.05. (b) The same kernels on the measured roofline.

4.4 Why: kernels, the roofline and host overhead

Answer: *The roofline explains the vLLM kernels. Extra kernels, and the CPU cost of launching them, explain the HF stack.*

Kernels. To test the mechanism directly, we time each kernel alone on the matrix shapes of one 7B decoder layer (Figure 5a). Marlin W4A16 is 3.2× faster than BF16 at $M = 1$, as P1 predicts for four times fewer bytes. Its gain then shrinks and ends at $M \approx 333$, because it hits the same BF16 compute roof (Figure 5b). Real kernels reach that roof gradually, so the crossover sits above the ridge point. FP8's compute roof is about three times higher (645 vs. 218 TFLOP/s), so FP8 is still 1.9× faster at $M = 8192$.

Why the two W8A8 formats disagree. The number format is not the cause: the GPU multiplies INT8 as fast as FP8 (650 TOPS vs. 645 TFLOP/s). LLM.int8() is slow because it multiplies the outlier columns of the activations separately, in 16 bits. Each multiply thus needs over a dozen small

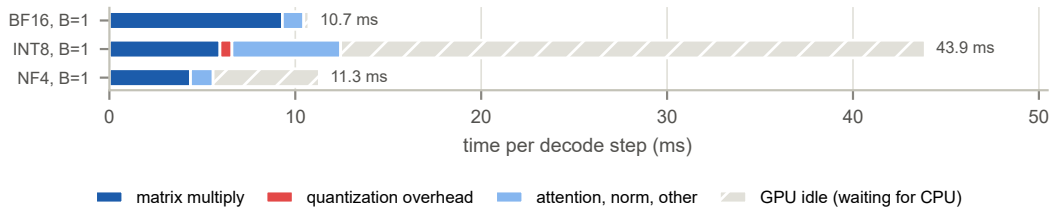


Figure 6: **Where an HF decode step goes** (Qwen2.5-7B, batch size 1, 512-token prompt): GPU time by kernel type, and the time the GPU waits for the CPU (hatched).

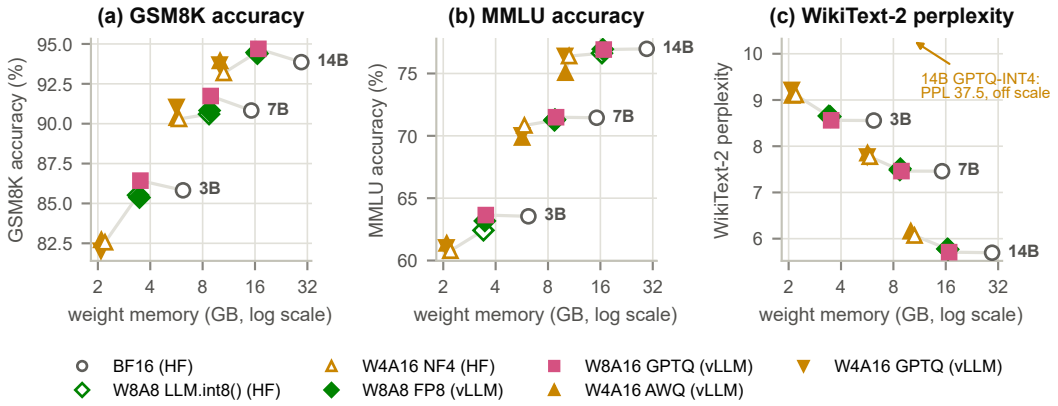


Figure 7: **Q4: quality against weight memory** for every Qwen2.5 configuration. Lines connect the configurations of one model size; colours and markers follow the quantization type (Table 1). A larger model in 4 bits sits above a smaller model in 16 bits, at less memory.

kernels, and the CPU waits for the GPU to find the outliers. Its layer runs at 0.49× of the BF16 speed at $M = 1$ and at most 1.33× at large M; FP8, with two kernels, is faster at every M (Figure 5a).

Host overhead. The kernels explain the vLLM results, but not the HF ones. NF4's kernel alone is 2.39× faster than BF16 at $M = 1$, yet the whole NF4 model is only 1.13× faster, so time must be lost outside the kernels. A profile of one decode step shows where (Figure 6). With BF16 the GPU is busy 97% of the step, with NF4 only 49%, and with LLM.int8(), which launches 3.7× as many kernels, only 28%. For the rest of the step the GPU waits for Python to launch the next kernel. LLM.int8()'s INT8 multiplies do take half the time of BF16's (4.8 vs. 9.3 ms per step), but the kernels around them lose this gain. More layers mean more launches, so this overhead grows with depth; this is why NF4 is slower than BF16 on the 36-layer 3B model. vLLM avoids most of it with CUDA graphs, which replay a decode step with very few launches.

4.5 Do the results hold for real traffic?

Answer: *Yes. The serving stack matters even more than the format.* A replay of 300 ShareGPT requests repeats the batch-size story (Table 7). With one client, decoding is memory-bound, so AWQ-INT4 gives the lowest TPOT (4.1 ms vs. 9.6 ms for BF16). With 64 clients it becomes compute-bound, so FP8 gives the highest throughput (5,388 vs. 3,537 tokens/s). The stack matters more: with the same 16 requests in flight and the same BF16 weights, vLLM delivers 5.3× the throughput of HF, whose static batches wait for their longest request. This gap comes from continuous batching [27] and CUDA graphs, not from quantization (the stacks are timed differently; Appendix C).

4.6 Q4: How much quality does quantization cost?

Answer: *8-bit formats keep quality. 4-bit formats cost a little, most on small models. The saved memory is therefore best spent on a larger model.*

Table 2: Quality cost of each format. First row: BF16 values (WikiText-2 perplexity; GSM8K and MMLU accuracy in %). Other rows: change against HF BF16 of the same model (accuracy in points). **Bold**: significant after Bonferroni correction over the 18 comparisons per benchmark (exact McNemar test). Last row: BF16 on vLLM, the noise floor.

			Qwen2.5-3B			Qwen2.5-7B			Qwen2.5-14B		
Format	Stack	Type	PPL	GSM8K	MMLU	PPL	GSM8K	MMLU	PPL	GSM8K	MMLU
BF16 (value)	HF	—	8.56	85.8	63.5	7.46	90.8	71.4	5.70	93.9	77.0
FP8	vLLM	W8A8	+0.09	−0.5	−0.4	+0.05	0.0	−0.1	+0.07	+0.5	0.0
LLM.int8()	HF	W8A8	+0.10	−0.3	**−1.1**	+0.03	−0.2	−0.2	+0.08	+0.6	−0.3
GPTQ-INT8	vLLM	W8A16	0.00	+0.6	+0.1	0.00	+0.9	0.0	+0.01	+0.8	−0.1
AWQ-INT4	vLLM	W4A16	+0.54	**−3.1**	**−2.2**	+0.39	−0.5	**−1.6**	+0.46	+0.1	**−1.9**
GPTQ-INT4	vLLM	W4A16	+0.66	**−3.8**	**−2.5**	+0.32	+0.2	**−1.4**	+31.77‡	−0.2	−0.6
NF4	HF	W4A16	+0.55	**−3.3**	**−2.8**	+0.31	−0.5	**−0.6**	+0.37	−0.7	−0.6
BF16 (noise)	vLLM	—	0.00	+0.2	+0.1	0.00	+0.4	0.0	–	–	–

‡Large only on raw text; on chat-formatted text this checkpoint matches the other 14B formats (Section 4.6).

Noise floor. The gains above matter only if quality holds, so we first measure the noise floor. BF16 on the two stacks disagrees on 19 of the 7B model's GSM8K answers, because different kernels round differently, yet the accuracies differ by only 0.4 points (MMLU: less than 0.1).

8 bits keep quality. Measured against this floor, 8-bit formats show no loss. GPTQ-INT8 and FP8 stay within 0.9 points of BF16 on both benchmarks for all three sizes; on GSM8K, the only 8-bit differences with $p < 0.05$ are improvements (GPTQ-INT8 on 7B, $+0.9$ points, $p = 0.043$; GPTQ-INT8 on 14B, $+0.8$ points, $p = 0.013$), and neither survives the Bonferroni correction (Table 2). The only measurable 8-bit cost is LLM.int8() on the 3B model (-1.1 MMLU points, $p < 0.001$).

4 bits cost a little. Removing four more bits does cost quality. The 4-bit formats change MMLU accuracy by -1.6 to -0.6 points on 7B and -1.9 to -0.6 on 14B. The cost is largest on the smallest model: -2.8 to -2.2 points on 3B, where GSM8K drops as well (-3.8 to -3.1). A small model has less spare capacity to absorb the rounding error.

More parameters beat more bits. This cost is smaller than the gain from a larger model, so at a fixed memory budget parameters are worth more than bits (Figure 7), as scaling laws predict [4]. Qwen2.5-14B in AWQ-INT4 needs 10.1 GB of weights, less than the 15.2 GB of Qwen2.5-7B in BF16, and scores 93.9% vs. 90.8% on GSM8K and 75.1% vs. 71.4% on MMLU (McNemar $p < 0.001$ for both).

One metric is not enough. The 14B GPTQ-INT4 checkpoint keeps its task accuracy (93.7% on GSM8K), yet its WikiText-2 perplexity is 37.5 instead of 5.71–6.16. It ranks the correct next token first for only 37% of raw WikiText tokens (other 14B formats: 60–62%), but for 77% of chat-formatted answers (others: 77%). The checkpoint still serves chat well but has lost much of its general language modeling, which any single metric would miss.

4.7 Do the results generalize? Qwen3-8B and Qwen3-32B

Answer: *The systems effects carry over to a newer model family. The quality order of the formats does not, so it must be measured for each model.*

Qwen3-8B. If the speed effects come from the hardware and the kernels, they should not depend on the model. Qwen3-8B tests this with a different shape: 36 layers and 8 KV heads, against 28 and 4 for Qwen2.5-7B. The systems effects carry over (Figure 8a, Appendix E): at batch size 1, AWQ-INT4 decodes 2.33× faster than BF16 (Qwen2.5-7B: 2.44×), FP8 1.54×, and LLM.int8() runs at 0.28× of the BF16 speed. The one change confirms the host-overhead explanation: with more layers, NF4 falls from 1.13× to 0.90× of the BF16 speed. Quality, in contrast, depends on the weights rather than on the kernels. Quality costs stay small (Figure 8b; no change above 0.7 points). But the order of the formats changes: FP8 costs 0.1 MMLU points on Qwen2.5-7B but 0.7 on Qwen3-8B, as much as AWQ-INT4. Which format keeps quality depends on the model.

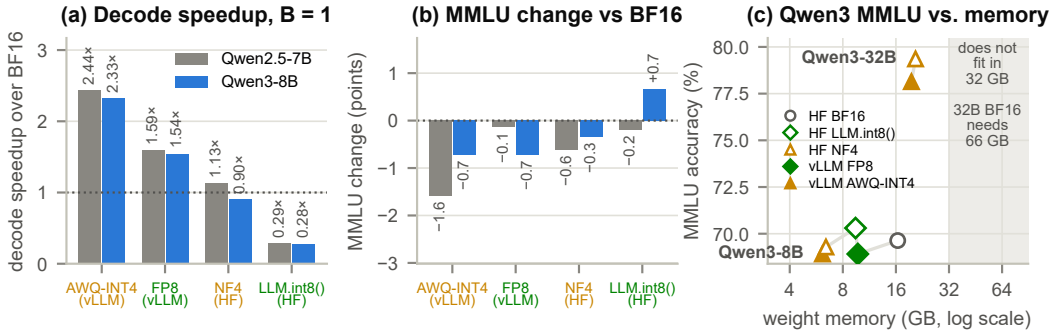


Figure 8: **Generalization.** (a) Decode speedup and (b) MMLU change against BF16 for Qwen2.5-7B and Qwen3-8B. (c) Qwen3 MMLU accuracy against weight memory: Qwen3-32B fits only in 4 bits, and it beats Qwen3-8B in BF16.

Table 3: A decision guide for quantized LLM inference on one GPU.

Situation	Choice	Evidence
Planning memory	Use the real ratio, not $16/b$	4-bit saves only 2.60×–2.70× (§4.1)
Few users ($B \lesssim 16$)	AWQ/GPTQ-INT4, fused kernels	2.44× faster decode (§4.2)
Many users ($B \gtrsim 64$)	FP8 W8A8	Faster at every B; 1.73× throughput (§4.3)
Tight memory budget	Larger model in 4 bits	4-bit 32B beats BF16 8B (§4.7)
Small model, accuracy first	8-bit (FP8, GPTQ-INT8)	4-bit costs up to 2.8 MMLU points (§4.6)
Only HF + bitsandbytes	Quantize for memory, not speed	LLM.int8() is 3.4× slower (§4.4)

Qwen3-32B. Spending memory on parameters matters most when a model would not fit otherwise. Qwen3-32B needs 66 GB in BF16 and about half of that in 8 bits, more than the GPU's 32 GB, so it runs here only in 4 bits (AWQ-INT4: 19.6 GB plus 29K cached tokens). Compared with Qwen3-8B in BF16 (16.4 GB), it costs 13.3 vs. 10.3 ms per token, but it scores 94.2% vs. 91.4% on GSM8K ($p < 0.001$) and 78.1% vs. 69.6% on MMLU ($p < 0.001$; Figure 8c). A model four times larger, using only 1.20× the weight memory, is significantly better on both benchmarks.

5 Discussion

A decision guide. Table 3 turns each link of the chain into a rule.

Lessons for system design. Weight-only formats save bandwidth but not compute, so to stay fast at large batches the math itself must run in low precision, as FP8 does here and FP4 can on the newest GPUs. Even then, the speedup reaches the user only through a fused kernel and a decode loop with few launches: LLM.int8() also multiplies in 8 bits, yet it is slower than BF16 everywhere. The software stack therefore matters as much as the format. Once the weights are in 4 bits, the KV cache is the largest 16-bit user of memory, so KV-cache quantization [11, 19] is the next step.

Limitations. We test one GPU and dense Qwen models, and crossover batch sizes depend on each GPU's ratio of compute to bandwidth. The GPU also drives a display, so its clocks were not locked. We do not study 4-bit activations, KV-cache quantization or mixture-of-experts models.

6 Conclusion

Quantization changes memory, speed and quality, and on one consumer GPU each change limits the next. 4-bit weights save only 2.60×–2.70×, and less at run time, because the embedding, the KV cache and the activations stay in 16 bits. The saved bytes become speed only while decoding is memory-bound and a fused kernel hides the dequantization (2.44× on vLLM, 1.13× with bitsandbytes). Batching ends that regime near batch size 175; after it, only FP8, whose fused kernel also multiplies in 8 bits, stays faster. The quality cost is small at 8 bits and 0.6–2.8 MMLU points at 4 bits, less than the gain from a larger model. Quantization therefore pays off most as a way to run a larger model: smaller is not always faster, but it can be both smaller and better.

Author contributions

Sijia Zhong built the harness and the memory study (Q1), Zheng Zheng ran the sweeps and the kernel and profiling analysis (Q2, Q3), and Haofei Sun ran the real-traffic and quality evaluations (Q4). All three analyzed the results and wrote the report and the slides.

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A Additional memory results

Figure 9 and Table 4 give the weight memory of every configuration and model size. Figure 10 splits the peak memory of the 7B model on HF into weights, KV cache and activations. It shows why the run-time saving shrinks: as the batch grows, the KV cache and the activations take a growing share of the peak memory, and neither is quantized. Figure 11 shows the KV-cache capacity that vLLM gets from the memory each format frees.

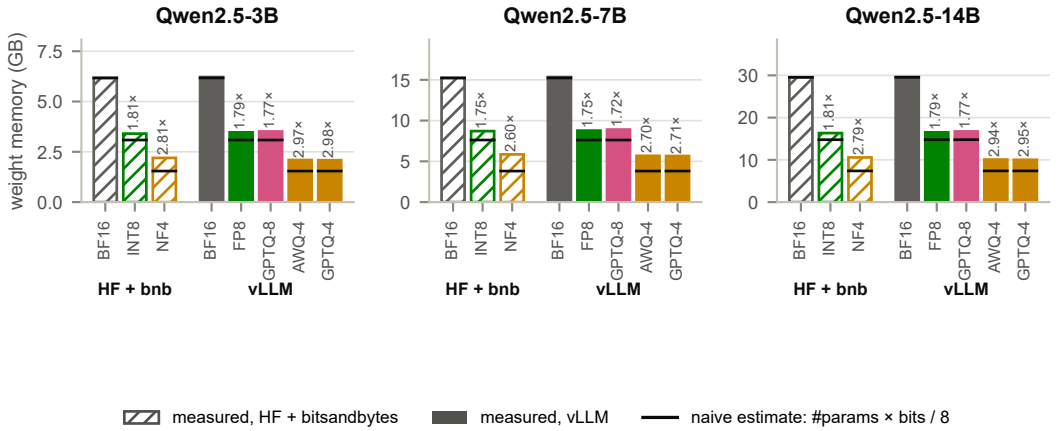


Figure 9: Weight memory of every configuration for the three Qwen2.5 sizes (bars: measured; black ticks: #params $\times$ bits/8). Labels give the saving against BF16 on the same backend.

Table 4: Weight memory: the naive estimate (#params$\times$bits/8), our model (embedding, LM head and norms stay in 16 bits; each format adds its scales), and the value measured on the GPU.

Model	Config	Naive (GB)	Model (GB)	Measured (GB)	Ratio vs BF16	Naive ratio
Qwen2.5-7B	BF16 (HF)	15.23	15.23	15.23	1.00×	1.00×
Qwen2.5-7B	BF16 (vLLM)	15.23	15.23	15.34	1.00×	1.00×
Qwen2.5-7B	W8A8 FP8 (vLLM)	7.62	8.71	8.77	1.75×	2.00×
Qwen2.5-7B	W8A8 LLM.int8() (HF)	7.62	8.71	8.71	1.75×	2.00×
Qwen2.5-7B	W8A16 GPTQ (vLLM)	7.62	8.86	8.90	1.72×	2.00×
Qwen2.5-7B	W4A16 AWQ (vLLM)	3.81	5.57	5.68	2.70×	4.00×
Qwen2.5-7B	W4A16 GPTQ (vLLM)	3.81	5.57	5.66	2.71×	4.00×
Qwen2.5-7B	W4A16 NF4 (HF)	3.81	5.85	5.85	2.60×	4.00×
Qwen2.5-14B	BF16 (HF)	29.54	29.54	29.54	1.00×	1.00×
Qwen2.5-14B	BF16 (vLLM)	29.54	29.54	29.60	1.00×	1.00×
Qwen2.5-14B	W8A8 FP8 (vLLM)	14.77	16.33	16.51	1.79×	2.00×
Qwen2.5-14B	W8A8 LLM.int8() (HF)	14.77	16.33	16.34	1.81×	2.00×
Qwen2.5-14B	W8A16 GPTQ (vLLM)	14.77	16.64	16.73	1.77×	2.00×
Qwen2.5-14B	W4A16 AWQ (vLLM)	7.39	9.98	10.07	2.94×	4.00×
Qwen2.5-14B	W4A16 GPTQ (vLLM)	7.39	9.98	10.02	2.95×	4.00×
Qwen2.5-14B	W4A16 NF4 (HF)	7.39	10.55	10.58	2.79×	4.00×

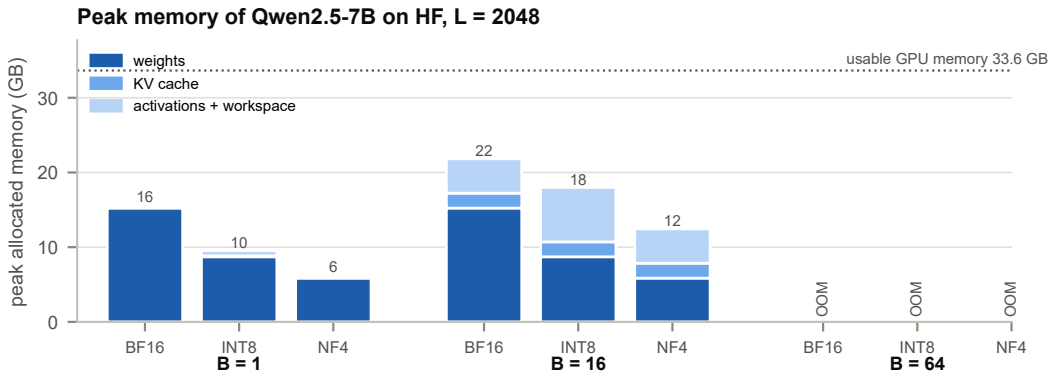


Figure 10: Peak allocated memory of Qwen2.5-7B on HF with 2048-token prompts, split into weights, KV cache and activations. OOM: out of memory.

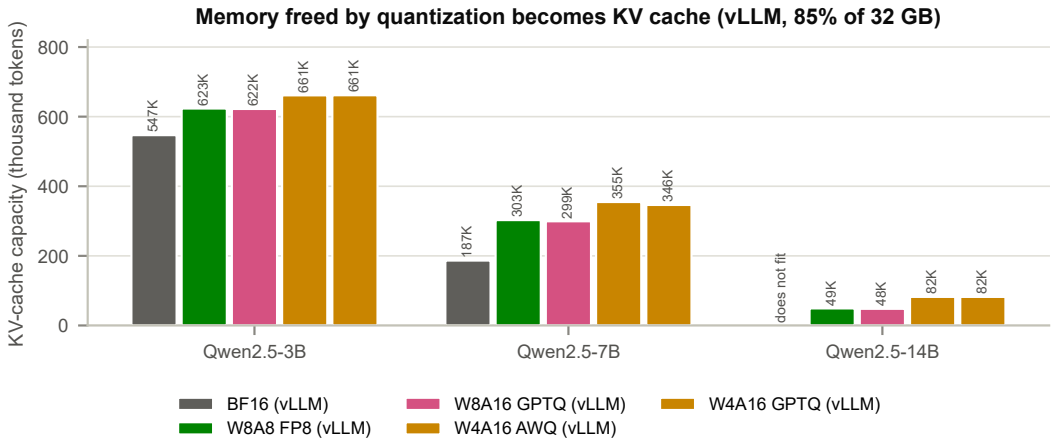


Figure 11: KV-cache capacity that vLLM allocates with 85% of the GPU after loading the weights.

B Additional speed results

Table 5 and Figure 12 give the absolute latency, throughput and energy behind Figure 3, and Table 6 lists where each format stops paying off (Figure 4). Figure 13 shows the energy per output token. GPU power stays close to the board limit in every configuration (410–579 W), so energy per token follows latency. LLM.int8() draws little power (164 W) but runs so long that it uses the most energy per token. Figure 14 shows how the answer length moves the speedup (Section 4.3).

Table 5: Latency, throughput and energy of Qwen2.5-7B with 512-token prompts and 128 output tokens (mean of 3 runs). Subscripts give the batch size. TTFT and TPOT in ms; Speedup_1 is the decode speedup over BF16 on the same backend; energy is GPU energy per output token.

Config	TTFT_1	TPOT_1	TPOT_{64}	tok/s_{64}	Speedup_1	mJ/tok_1	mJ/tok_{64}
BF16 (HF)	38.4	10.90	20.04	1669	1.00×	5092	294
BF16 (vLLM)	37.0	9.52	12.39	2209	1.00×	4878	249
W8A8 FP8 (vLLM)	22.8	5.99	7.88	3886	1.59×	2507	141
W8A8 LLM.int8() (HF)	57.8	37.45	47.14	1003	0.29×	6187	305
W8A16 GPTQ (vLLM)	38.5	5.80	8.41	2471	1.64×	2588	234
W4A16 AWQ (vLLM)	38.6	3.90	7.99	2544	2.44×	1919	224
W4A16 GPTQ (vLLM)	36.9	3.88	7.89	2642	2.45×	1840	219
W4A16 NF4 (HF)	48.3	9.62	22.87	1550	1.13×	3188	340

Table 6: Where each format stops paying off (Qwen2.5-7B, 512-token prompts, 128 output tokens). Speedup over BF16 on the same backend at batch size 1 and at the largest batch we ran (in brackets), and the batch size up to which the format is faster (linear interpolation in $\log B$).

	Decode step (TPOT)			Whole request (E2E)		
Config	B=1	largest B	faster up to	B=1	largest B	faster up to
W8A8 FP8 (vLLM)	1.59×	1.36× (256)	all B	1.59×	1.73× (256)	all B
W8A8 LLM.int8() (HF)	0.29×	0.43× (64)	never	0.30×	0.60× (64)	never
W8A16 GPTQ (vLLM)	1.64×	1.00× (256)	$B \approx 251$	1.61×	0.97× (256)	$B \approx 123$
W4A16 AWQ (vLLM)	2.44×	1.02× (256)	all B	2.33×	0.98× (256)	$B \approx 175$
W4A16 GPTQ (vLLM)	2.45×	1.05× (256)	all B	2.35×	1.03× (256)	all B
W4A16 NF4 (HF)	1.13×	0.88× (64)	$B \approx 7$	1.12×	0.93× (64)	$B \approx 7$

Qwen2.5-3B. Figures 15 and 16 repeat the latency and crossover analysis for the 3B model. On vLLM the picture is the same as for 7B. On HF the smaller model is worse off. Its BF16 step is already limited by the host (8.3 ms per step against a memory floor of 3.8 ms), so fewer bytes cannot help: NF4 never beats BF16, and LLM.int8() falls to 0.13× of the BF16 speed, because the 36-layer model launches more kernels per step than the 28-layer 7B model.

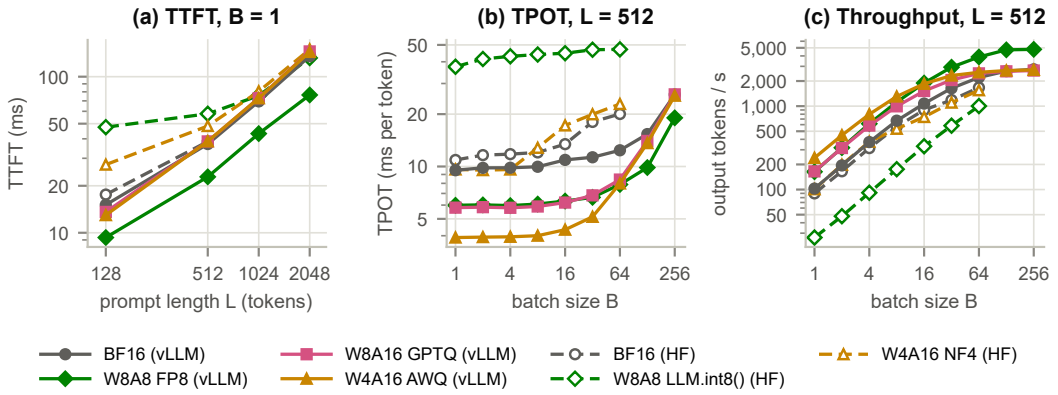


Figure 12: Absolute latency and throughput of Qwen2.5-7B with 128 generated tokens: (a) TTFT of one request, (b) TPOT and (c) output throughput for 512-token prompts. Figure 3 shows the same data as speedups over BF16.

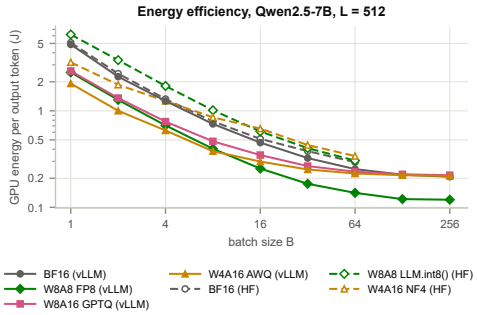


Figure 13: GPU energy per output token (NVML), Qwen2.5-7B, 512-token prompts.

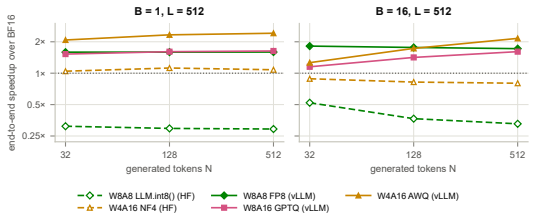


Figure 14: End-to-end speedup over BF16 against the number of generated tokens (Qwen2.5-7B, 512-token prompts). Longer answers raise the share of decode, and with it the gain of weight-only formats.

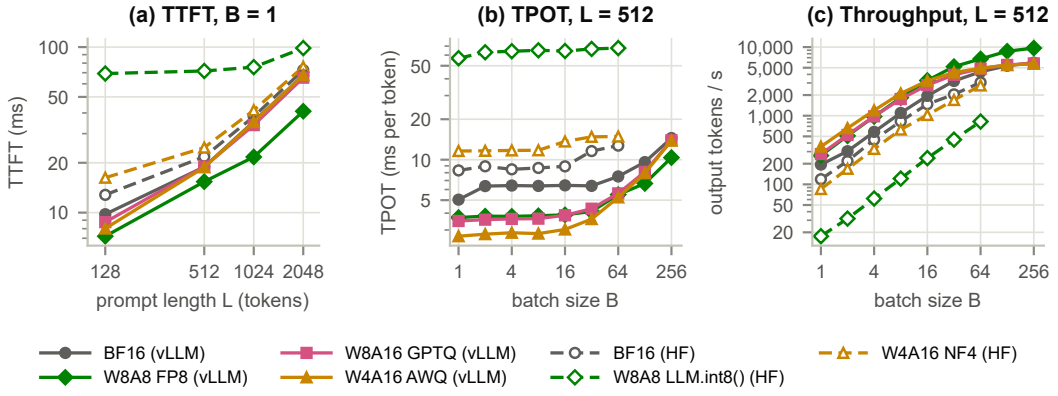


Figure 15: Latency and throughput of Qwen2.5-3B (same layout as Figure 12).

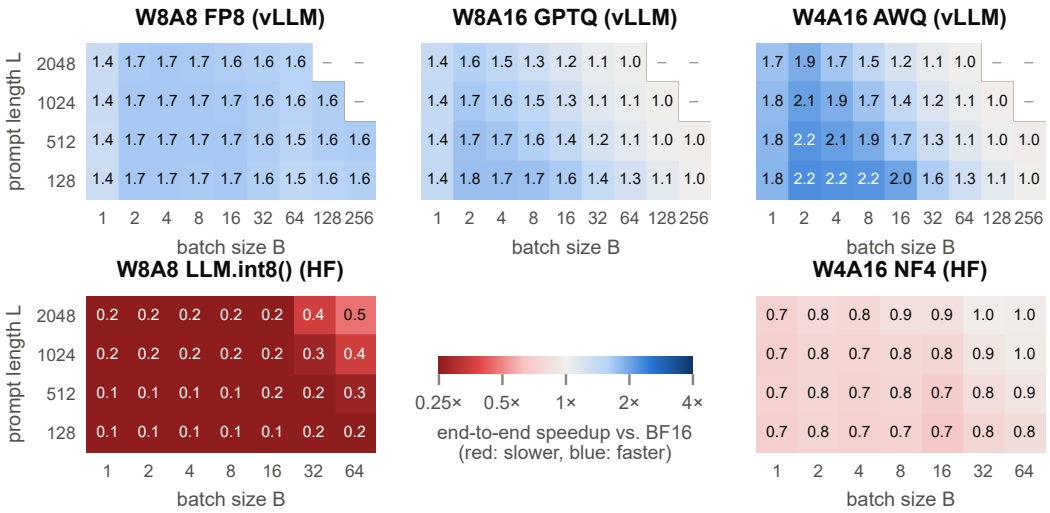


Figure 16: End-to-end speedup over BF16 for Qwen2.5-3B (same layout as Figure 4).

C Real-traffic details

Table 7 summarizes the ShareGPT replay of Section 4.5. The HF and vLLM numbers compare deployment stacks, not formats, and are timed differently. vLLM is measured by `vllm bench serve` at the client, after 8 warm-up requests; its TTFT includes the server's queueing and HTTP overhead. HF has no warm-up, and its TTFT is the time from the start of a batch's `generate` call to its first token; it leaves out the time a request waits for earlier batches, which favors HF. Even so, HF's median TTFT is 797 ms against 34 ms for vLLM. Figure 17 shows the lengths of the 300 requests, and Figure 18 and Table 8 give the full serving results, including the 99th-percentile latencies.

Table 7: Real traffic: 300 ShareGPT requests replayed against Qwen2.5-7B with their reference answer lengths. C is the number of concurrent requests (HF runs static batches of 16). Output tokens per second, and median TTFT and TPOT in ms.

	$C = 1$		$C = 16$		$C = 64$	
Config	tok/s	TPOT	tok/s	TTFT	tok/s	TPOT
BF16 (HF) (static)	–	–	254	797	–	–
BF16 (vLLM)	104	9.6	1,334	34	3,537	13.6
FP8 (vLLM)	164	6.1	2,237	21	5,388	8.9
LLM.int8() (HF) (static)	–	–	119	691	–	–
GPTQ-8 (vLLM)	166	6.0	2,166	22	4,733	10.5
AWQ-4 (vLLM)	240	4.1	3,020	18	5,180	10.1
GPTQ-4 (vLLM)	241	4.1	2,970	18	5,181	10.0
NF4 (HF) (static)	–	–	215	842	–	–

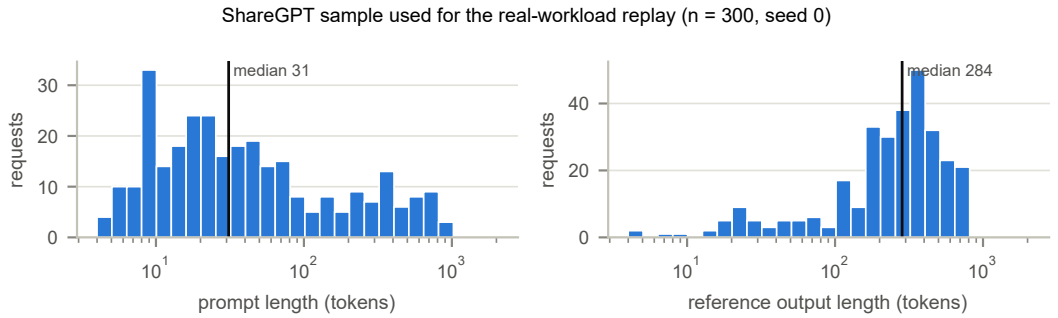


Figure 17: Prompt and answer lengths of the 300 ShareGPT requests used in the replay.

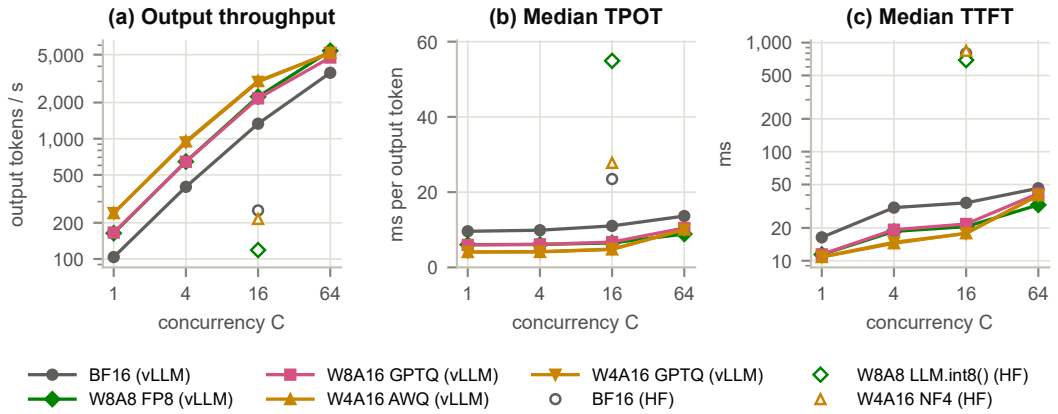


Figure 18: The ShareGPT replay against Qwen2.5-7B: vLLM at 1 to 64 concurrent clients (lines) and HF with static batches of 16 (hollow markers). TTFT and TPOT are medians over requests.

Table 8: All serving results: 300 ShareGPT requests replayed against Qwen2.5-7B (reference answer lengths). C: concurrent clients for vLLM, static batch size for HF. Throughput in requests/s and output tokens/s; TTFT and TPOT in ms (median / 99th percentile).

Config	C	req/s	tok/s	TTFT p50 / p99	TPOT p50 / p99
BF16 (HF) (static)	16	0.84	254	797 / 1244	23.5 / 29.7
BF16 (vLLM)	1	0.35	104	16 / 62	9.6 / 9.6
BF16 (vLLM)	64	11.76	3537	46 / 1654	13.6 / 16.7
FP8 (vLLM)	1	0.54	164	11 / 44	6.1 / 6.2
FP8 (vLLM)	64	17.92	5388	33 / 1485	8.9 / 15.1
LLM.int8() (HF) (static)	16	0.39	119	691 / 1158	54.9 / 64.1
GPTQ-8 (vLLM)	1	0.55	166	11 / 63	6.0 / 6.2
GPTQ-8 (vLLM)	64	15.74	4733	41 / 1743	10.5 / 18.0
AWQ-4 (vLLM)	1	0.80	240	11 / 65	4.1 / 4.5
AWQ-4 (vLLM)	64	17.23	5180	40 / 1747	10.1 / 14.8
GPTQ-4 (vLLM)	1	0.80	241	11 / 62	4.1 / 4.4
GPTQ-4 (vLLM)	64	17.23	5181	40 / 1722	10.0 / 19.3
NF4 (HF) (static)	16	0.72	215	842 / 1271	27.8 / 35.4

D Full quality results

Table 9 lists every quality number with its p-value, for all five models.

Table 9: All quality results. WikiText-2 perplexity (2048-token windows), GSM8K accuracy (%, 1,319 questions, zero-shot chain of thought, greedy) and MMLU accuracy (%, 14,042 questions, zero-shot). Δ is the change against HF BF16 of the same model (Qwen3-32B has no BF16 run: it does not fit the GPU); p is the two-sided exact McNemar test on the paired answers.

Model	Config	PPL	ΔPPL	GSM8K	Δ	p	MMLU	Δ	p
Qwen2.5-3B	BF16 (HF)	8.56	–	85.8	–	–	63.5	–	–
	BF16 (vLLM)	8.56	0.00	86.0	+0.2	0.871	63.7	+0.1	0.075
	FP8 (vLLM)	8.64	+0.09	85.4	−0.5	0.610	63.2	−0.4	0.038
	LLM.int8() (HF)	8.66	+0.10	85.5	−0.3	0.777	62.4	−1.1	<0.001
	GPTQ-8 (vLLM)	8.56	0.00	86.4	+0.6	0.322	63.6	+0.1	0.261
	AWQ-4 (vLLM)	9.10	+0.54	82.7	−3.1	0.002	61.4	−2.2	<0.001
	GPTQ-4 (vLLM)	9.22	+0.66	82.0	−3.8	<0.001	61.1	−2.5	<0.001
	NF4 (HF)	9.11	+0.55	82.6	−3.3	<0.001	60.8	−2.8	<0.001
Qwen2.5-7B	BF16 (HF)	7.46	–	90.8	–	–	71.4	–	–
	BF16 (vLLM)	7.46	0.00	91.2	+0.4	0.359	71.4	0.0	0.704
	FP8 (vLLM)	7.51	+0.05	90.8	0.0	1.000	71.3	−0.1	0.327
	LLM.int8() (HF)	7.49	+0.03	90.6	−0.2	0.771	71.2	−0.2	0.134
	GPTQ-8 (vLLM)	7.46	0.00	91.7	+0.9	0.043	71.5	0.0	0.610
	AWQ-4 (vLLM)	7.85	+0.39	90.4	−0.5	0.576	69.9	−1.6	<0.001
	GPTQ-4 (vLLM)	7.78	+0.32	91.1	+0.2	0.828	70.0	−1.4	<0.001
	NF4 (HF)	7.77	+0.31	90.3	−0.5	0.500	70.8	−0.6	0.002
Qwen2.5-14B	BF16 (HF)	5.70	–	93.9	–	–	77.0	–	–
	FP8 (vLLM)	5.77	+0.07	94.4	+0.5	0.371	76.9	0.0	0.862
	LLM.int8() (HF)	5.78	+0.08	94.5	+0.6	0.280	76.6	−0.3	0.025
	GPTQ-8 (vLLM)	5.71	+0.01	94.7	+0.8	0.013	76.9	−0.1	0.416
	AWQ-4 (vLLM)	6.16	+0.46	93.9	+0.1	1.000	75.1	−1.9	<0.001
	GPTQ-4 (vLLM)	37.46[‡]	+31.77	93.7	−0.2	0.899	76.4	−0.6	0.005
	NF4 (HF)	6.07	+0.37	93.2	−0.7	0.281	76.4	−0.6	0.003
Qwen3-8B	BF16 (HF)	9.73	–	91.4	–	–	69.6	–	–
	BF16 (vLLM)	9.72	0.00	92.0	+0.6	0.215	69.6	0.0	0.610
	FP8 (vLLM)	9.73	0.00	91.8	+0.5	0.488	68.9	−0.7	<0.001
	LLM.int8() (HF)	9.72	0.00	92.0	+0.6	0.382	70.3	+0.7	<0.001
	AWQ-4 (vLLM)	10.18	+0.46	91.1	−0.2	0.818	68.9	−0.7	0.003
	NF4 (HF)	9.98	+0.25	91.5	+0.2	0.909	69.3	−0.3	0.156
Qwen3-32B	AWQ-4 (vLLM)	7.88[§]	–	94.2	–	–	78.1	–	–
	NF4 (HF)	7.93	–	92.6	–	–	79.4	–	–

[‡]High only on raw text, not on chat-formatted text (Section 4.6). [§]Run with BF16 activations: with the checkpoint's own FP16 activations, some 2048-token windows overflow and the loss becomes NaN.

E Generalization details

Table 10 compares Qwen2.5-7B and Qwen3-8B on the same grid.

Table 10: The main results for Qwen2.5-7B and Qwen3-8B (non-thinking mode) on the same grid (512-token prompts, 128 output tokens). Throughput is compared at the largest batch at which both models' KV caches hold every request ($B = 64$). Qwen3-8B has no official GPTQ checkpoint.

Metric (vs. BF16)	Qwen2.5-7B	Qwen3-8B
Weight saving, AWQ-INT4 (vLLM)	2.70×	2.67×
Weight saving, NF4 (HF)	2.60×	2.55×
Decode speedup $B = 1$, AWQ-INT4	2.44×	2.33×
Decode speedup $B = 1$, FP8	1.59×	1.54×
Decode speedup $B = 1$, NF4	1.13×	0.90×
Decode speedup $B = 1$, LLM.int8()	0.29×	0.28×
Throughput $B = 64$, AWQ-INT4	1.15×	1.12×
Throughput $B = 64$, FP8	1.76×	1.67×
GSM8K Δ pp (p), FP8	0.0 (1.000)	+0.5 (0.488)
GSM8K Δ pp (p), LLM.int8()	−0.2 (0.771)	+0.6 (0.382)
GSM8K Δ pp (p), AWQ-INT4	−0.5 (0.576)	−0.2 (0.818)
GSM8K Δ pp (p), NF4	−0.5 (0.500)	+0.2 (0.909)
MMLU Δ pp (p), FP8	−0.1 (0.327)	−0.7 (<0.001)
MMLU Δ pp (p), LLM.int8()	−0.2 (0.134)	+0.7 (<0.001)
MMLU Δ pp (p), AWQ-INT4	−1.6 (<0.001)	−0.7 (0.003)
MMLU Δ pp (p), NF4	−0.6 (0.002)	−0.3 (0.156)

F Reproducibility

Every number in this report is computed from the raw measurements by `scripts/make_numbers.py`, and the figures and tables come from `make_figures.py` and `make_tables.py`. `run_all.sh` runs all experiments again (about twelve GPU-hours on an RTX 5090). A job queue ran the experiments one at a time, and only while no other user's process was on the GPU. Rows written while the GPU was shared (2 windows, 26 minutes in total, logged in `results/excluded_windows.jsonl`) were discarded and those jobs were run again. Re-measuring vLLM BF16 on the 7B model after the interruption reproduced the original TPOT within 0.5%.